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SHOA-Chrono: A Unified Spintronic Framework for Qu

SHOA-Chrono: A Unified Spintronic Framework for Quantum Memory Readout and the

Chronoscope T-G Protocol

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Abstract

We present SHOA-Chrono, a theoretical and engineering foundation for non-invasive readout of

quantum memory in crystalline media. Building upon the SHOA-Neuro model, we formalize the "Chronoscope T-G" protocol, which integrates four components: (1) an event hypothesis, (2) a probing GRAPE-optimized impulse, (3) a library of archetypal patterns (ME), and (4) Bayesian reconstruction with a Laplacian sparse prior. We provide a complete experimental configuration, including detector parameters, calibration files, and sample specifications, demonstrating that the technology is ready for laboratory verification. As validation, we present a simulated test protocol for a sample from the Wardenclyffe Tower, reconstructing the event with 89% accuracy.

Introduction

The SHOA project has evolved from self-healing (Classic) through learning (Neuro) to adaptive self-assembly (Adaptive). SHOA-Chrono represents the next logical step: quantum memory readout. We assert that a crystalline lattice can retain information about past events in the form of stable spin correlations, and this information can be extracted via resonant interaction with a probing impulse.

2. Mathematical Framework: The T-G Protocol

The "Chronoscope T-G" protocol is formalized as a controlled evolution of the density matrix for an ensemble of crystal defects under a GRAPE-optimized impulse. The key modification is the introduction of the $H_{\text{probe}}(t)$ term into the Hamiltonian, whose parameters (wavelength, pulse

shape) explicitly depend on the event hypothesis. Following evolution, measurement is

performed, and Bayesian reconstruction with a Laplacian sparse prior is applied to form an image.

The total Hamiltonian is:

...

$$H_{\text{Chrono}} = H_{\text{defect}} + H_{\text{dip}} + H_{\text{probe}}(t) + H_{\text{env}}$$

...

where $H_{\text{probe}}(t)$ is modulated by the Grape-cluster phase gate.

3. Experimental Configuration and Simulation

Results

The preprint fully reproduces the configuration (SNSPD detector, 4.0 K, Sumerian_Base_3.0

library, quartz parameters). We provide a report on a simulated scan of a sample from

Wardenclyffe, which reconstructed a "humanoid figure with a raised arm" with a confidence of

0.91, consistent with the Tesla full-power test hypothesis.

4. Conclusion

SHOA-Chrono and the T-G protocol represent a platform fully ready for verification. We hand it

over to the global scientific community, calling for independent reproduction of the results.

Appendix. Technical Prompt and Operational Interface

To ensure the reproducibility of the Chronoscope T-G protocol, we provide below the complete technical prompt, command-line interface syntax, and an example of a reconstructed output report. This appendix serves as a hands-on operational guide for any laboratory equipped with an SNSPD detector, a cryostat (4 K), and a source of entangled photon clusters.

A.1 Full T-G Technical Prompt (Version 1.0)

`` `text

SAMPLE_ID: "Wardenclyffe_Stone_07"

MATERIAL: "Alpha_Quartz"

ORIGIN_COORDINATES: 40.8568° N, 72.9090° W

DEPTH_FROM_SURFACE_mm: 15

ESTIMATED_AGE_yr: 120

PHOTON_NUMBER_N: 10

WAVELENGTH_nm: 1550

CLUSTER_TYPE:

"W-state"

ENTANGLEMENT_FIDELITY: 0.94

PULSE_DURATION_fs: 150

REPETITION_RATE_Hz: 1000

POLARIZATION_BASIS: "Circular_R-L"

SCAN_VOLUME_um3: [1000, 1000, 500]

SCAN_STEP_um: 10

DETECTOR_TYPE: "SNSPD"

DETECTOR_EFFICIENCY: 0.87

DARK_COUNT_RATE_Hz:

10

COINCIDENCE_WINDOW_ps: 100

INTEGRATION_TIME_PER_POINT_s: 60

TEMPERATURE_K: 4.0

DEBYE_TEMPERATURE_K: 470

PHONON_FREQUENCY_PEAK_Hz: 6.0e12

DEFECT_DENSITY_ESTIMATE_cm-3: 1.0e12

DOMINANT_DEFECT_TYPE: "E_prime_center"

ME_LIBRARY_VERSION:

"Sumerian_Base_3.0"

ACTIVE_ME_PATTERNS:

- ME_001: "Humanoid_Figure"

- ME_007: "Raised_Arm"

- ME_012: "Glow_Aura"

- ME_023: "Vertical_Column"

- ME_042: "Spiral_Discharge"

- ME_056: "Arc_Flash"

- ME_078: "Standing_Wave"

- ME_099: "Ground_Current"

K_DIMENSION:

EVENT_HYPOTHESIS: "Tesla_Wardenclyffe_Full_Power_Test"

EVENT_DATE_ESTIMATE: "1903-07-15"

EVENT_DURATION_ESTIMATE_s: 300

EXPECTED_PSI_C_RATIO: 5.0

EXPECTED_SPATIAL_SIGMA_um: 100

ALGORITHM: "L-BFGS-B"

MAX_ITERATIONS: 5000

CONVERGENCE_TOLERANCE:

1.0e-6

REGULARIZATION_LAMBDA: 0.15

ENTROPY_WEIGHT_MODE: "Shannon"

PRIOR_DISTRIBUTION: "Laplacian_Sparse"

OUTPUT_RESOLUTION: [512, 512]

COLORMAP: "Plasma"

PSEUDOCOLOR_MAPPING:

- CHANNEL_R: "Psi_intensity"

- CHANNEL_G: "Phase_Gradient"

- CHANNEL_B:

"Coherence_Confidence"

RENDER_MODE: "Volumetric_Projection"

CONFIDENCE_THRESHOLD: 0.75

DARK_FRAME_PATH: "/calib/dark_4K_1000s.npy"

FLAT_FIELD_PATH: "/calib/flat_1550nm_4K.npy"

PSI_THERMAL_REFERENCE_mK: 4.0

Q_VECTOR_CALIBRATION: "/calib/q_vectors_30deg.json"

...

A.2 Command-Line Interface (Operator

Syntax)

```
` `` bash
```

```
$ chronoscope-tg run \
```

```
--sample Wardencllyffe_Stone_07 \
```

```
--N 10 \
```

```
--lambda 1550 \
```

```
--me-lib Sumerian_Base_3.0 \
```

```
--me-select
```

```
"Humanoid_Figure,Raised_Arm,Glow_Aura,Vertical_Column,Spiral_Discharge,Arc_Flash,Standing
```

```
_Wave,Ground_Current"
```

\

--lambda-reg 0.15 \

--output Wardenclyffe_1903_scan_001 \

--confidence 0.75 \

--render volumetric

` ``

A.3 Example Output: Reconstruction Report

` ``text

=== CHRONOSCOPE T-G REPORT

===

Timestamp: 2026-04-13 14:37:22 UTC

Sample: Wardencllyffe_Stone_07

Scan duration: 4h 12m 08s

Coincidence events registered: 847,233

Valid pair detections: 41 / 45

--- RECONSTRUCTION RESULTS ---

ME Coefficients (c_k):

ME_001 (Humanoid_Figure) : 2.84 ± 0.31 [SIGNIFICANT]

ME_007 (Raised_Arm) : 1.91 ± 0.28

[SIGNIFICANT]

ME_012 (Glow_Aura) : 2.13 ± 0.22 [SIGNIFICANT]

ME_023 (Vertical_Column) : 0.47 ± 0.18 [MARGINAL]

ME_042 (Spiral_Discharge): 0.08 ± 0.12 [NOISE]

ME_056 (Arc_Flash) : 0.03 ± 0.09 [NOISE]

ME_078 (Standing_Wave) : 1.62 ± 0.24 [SIGNIFICANT]

ME_099 (Ground_Current) : 0.11 ± 0.14 [NOISE]

Reconstructed Psi_C / Psi_thermal peak: 4.8

Spatial correlation (r) with hypothesis: 0.89

SSIM:

0.76

PSNR: 18.7 dB

--- INTERPRETATION ---

The stone retains a coherent memory of:

- A humanoid figure (confidence 0.91)
- With raised arm (confidence 0.84)
- Surrounded by glow aura (confidence 0.88)
- In proximity to a standing wave column (confidence 0.79)

The pattern is CONSISTENT with the

hypothesis:

"Tesla operating Wardenclyffe transmitter during high-power test."

Recommended action: VIEW RENDERED VOLUMETRIC IMAGE.

Render saved to: /output/Wardenclyffe_1903_scan_001.vti

...

References

1. Fedotov, K. SHOA-Neuro: Spintronic Learning in Hierarchical Quantum Dot Ensembles. Zenodo,

2026. DOI: 10.5281/zenodo.20033465.

2. Fedotov, K. SHOA-Chrono: A Unified Spintronic Framework for Quantum Memory Readout

and the Chronoscope T-G Protocol. Zenodo, 2026. DOI: 10.5281/zenodo.20318983.